

Name: _____

Date: _____

ANGLES AND GEOMETRY, (ANGLES ON A STRAIGHT LINE AND AT A POINT)

LO: I can use the facts that angles on a straight line sum to 180° and angles at a point sum to 360° .

Angle rules

Angles on a **straight line** add up to **180°** . Angles at a **point** add up to **360°** .

Vertically opposite: When two lines cross, the opposite angles are equal.

Key facts

●	Angles on a straight line = 180°	half turn
●	Angles at a point = 360°	full turn
●	Vertically opposite angles are equal	X-shape
●	Angles on a line sum to 360°	should be 180°
●	Vertically opposite angles sum to 180°	they are equal

Key idea

Straight line = 180° . Full turn = 360° . Vertically opposite = equal.

$$a + b = 180^\circ \text{ (on a line);}$$

$$a + b + c + d = 360^\circ \text{ (at a point)}$$

Two fundamental angle facts

MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

Example 1 - angles on a straight line

Angles of 110° and x are on a straight line. Find x .

Angles on a straight line = 180°

$$110 + x = 180$$

$$x = 180 - 110 = 70^\circ$$

$$x = 70^\circ$$

Subtract the known angle from 180° .

Example 2 - three angles on a line

Three angles on a straight line are 40° , 85° and x . Find x .

$$40 + 85 + x = 180$$

$$125 + x = 180$$

$$x = 55^\circ$$

$$x = 55^\circ$$

Add the known angles, then subtract from 180° .

Example 3 - angles at a point

Three angles at a point are 120° , 150° and x . Find x .

Angles at a point = 360°

$$120 + 150 + x = 360$$

$$x = 360 - 270 = 90^\circ$$

$$x = 90^\circ$$

Subtract total of known angles from 360° .

Example 4 - vertically opposite

Two lines cross. One angle is 65° . Find all four angles.

Vertically opposite to 65° is also 65°

Adjacent angle: $180 - 65 = 115^\circ$

The four angles: 65° , 115° , 65° , 115°

$$65^\circ, 115^\circ, 65^\circ, 115^\circ$$

Opposite angles equal; adjacent angles sum to 180° .

YOUR TURN – PRACTICE (deliberate practice)

1. Two angles on a straight line

Find x .

- a) x and 130°
- b) x and 45°
- c) x and 90°
- d) x and 167°

2. Three angles on a line

Find x .

- a) 50° , 60° , x
- b) 35° , 90° , x
- c) 40° , 40° , x
- d) 25° , 75° , x

3. Angles at a point

Find x .

- a) 90° , 120° , 80° , x
- b) 100° , 100° , 100° , x
- c) 45° , 135° , 90° , x
- d) 150° , 70° , x

4. Vertically opposite

Two lines cross. One angle is given. Find the other three.

- a) 40°
- b) 72°
- c) 115°
- d) 90°

5. Mixed problems

Find x . State which rule you used.

- a) $2x$ and $3x$ on a straight line
- b) x , $2x$, $3x$ at a point
- c) $4x - 10$ and $2x + 40$ on a line
- d) Vertically opposite: $3x + 5 = 50$
- e) $x + 20$, $2x + 10$, $x + 30$ on a line

COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick () the ones that are correct. If it is wrong, write the correct answer.

a) Angles on a straight line add to 180° ☐

Correct answer: _____

Reason: _____

b) Angles at a point add to 180° ☐

Correct answer: _____

Reason: _____

c) Vertically opposite angles are equal ☐

Correct answer: _____

Reason: _____

d) A straight line has an angle of 360° ☐

Correct answer: _____

Reason: _____

e) When two lines cross, all four angles are equal ☐

Correct answer: _____

Reason: _____

6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) Three angles on a straight line are x° , $(2x + 10)^\circ$ and $(x + 30)^\circ$. Find each angle.

Expression: _____

Simplified: _____

b) Five angles meet at a point. Four of them are 80° , 95° , 55° and x° . The fifth angle is twice x . Find all five angles.

Expression: _____

Simplified: _____

TOP TIPS

Read the question carefully. Show all your working step by step.
Check your answer makes sense.

CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

